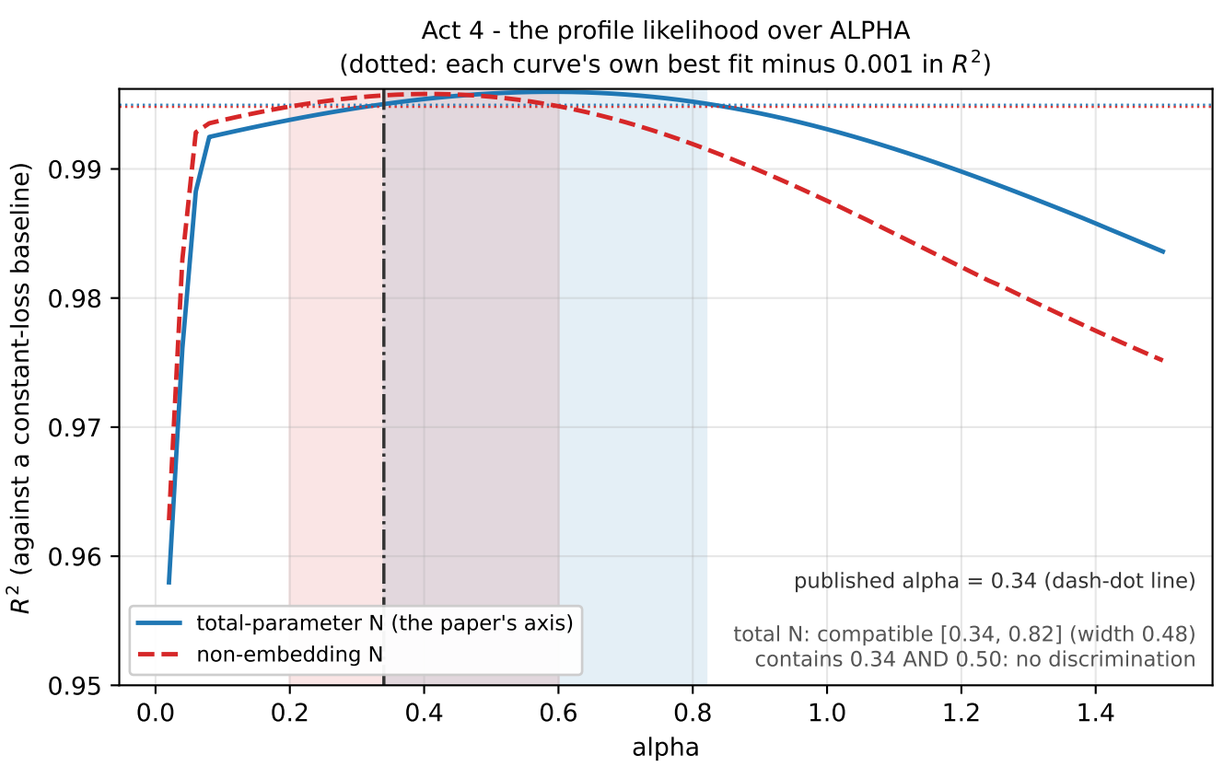
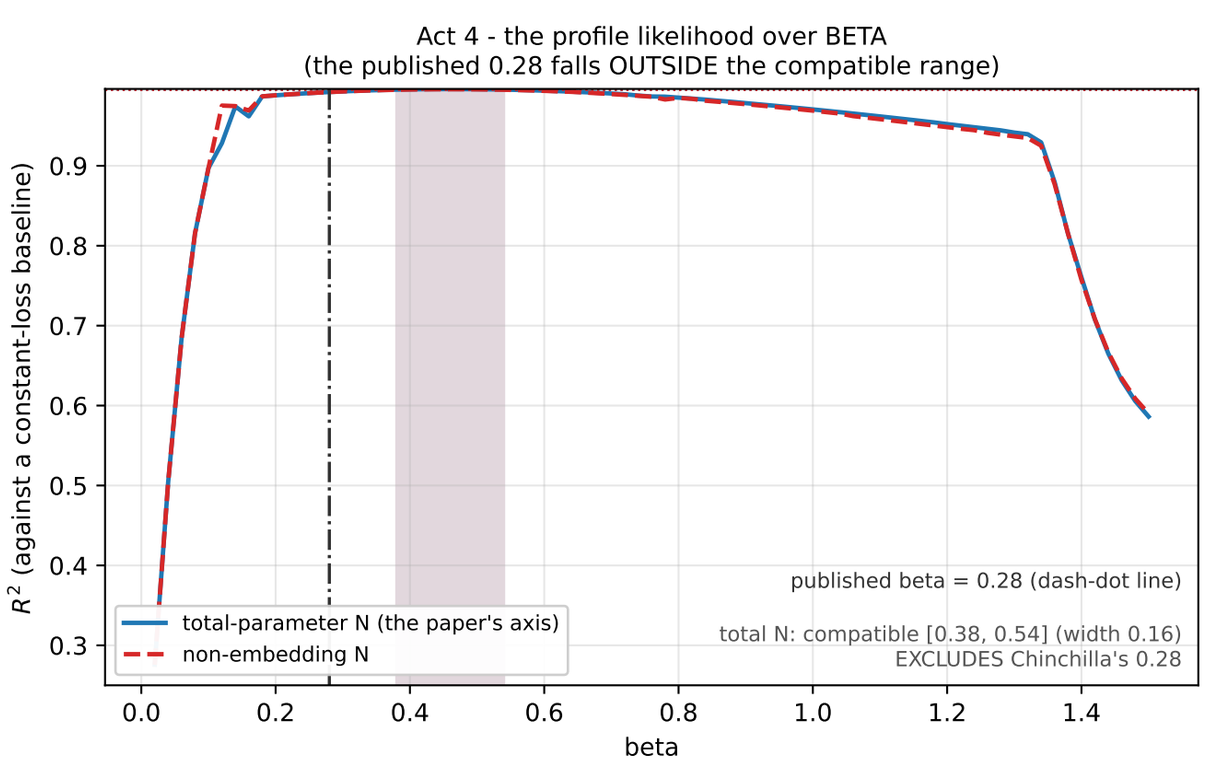
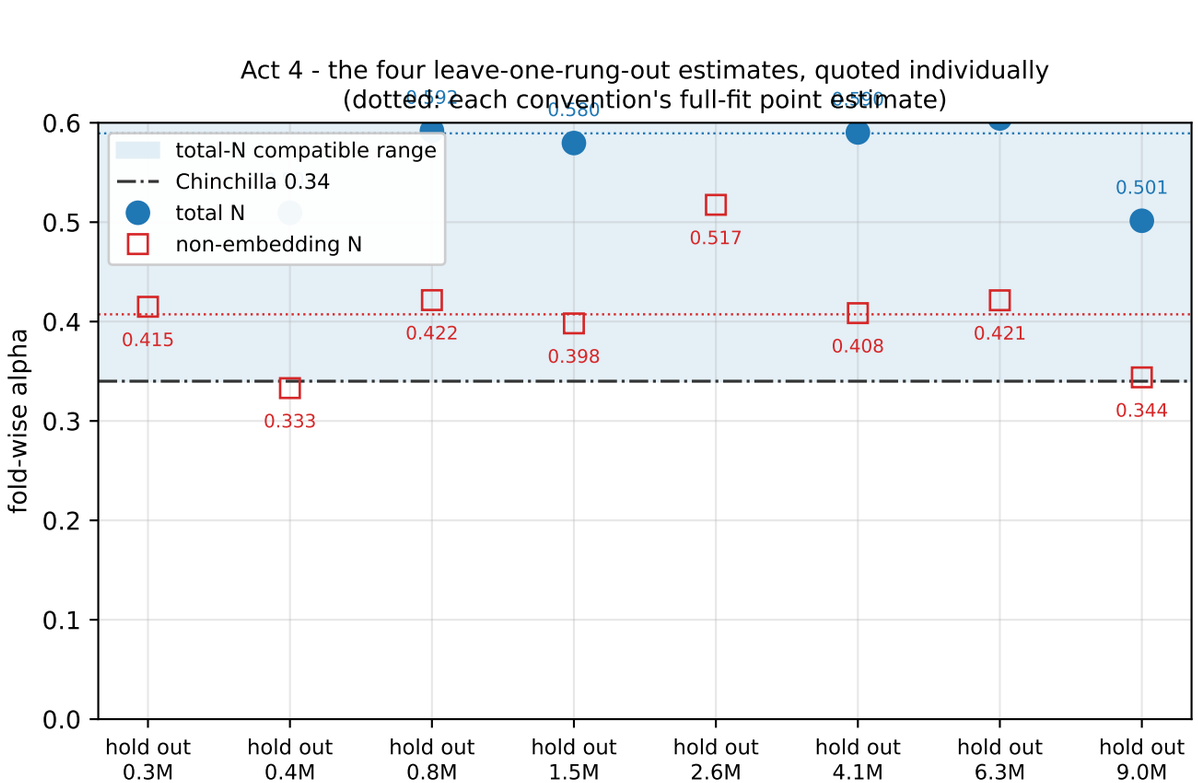
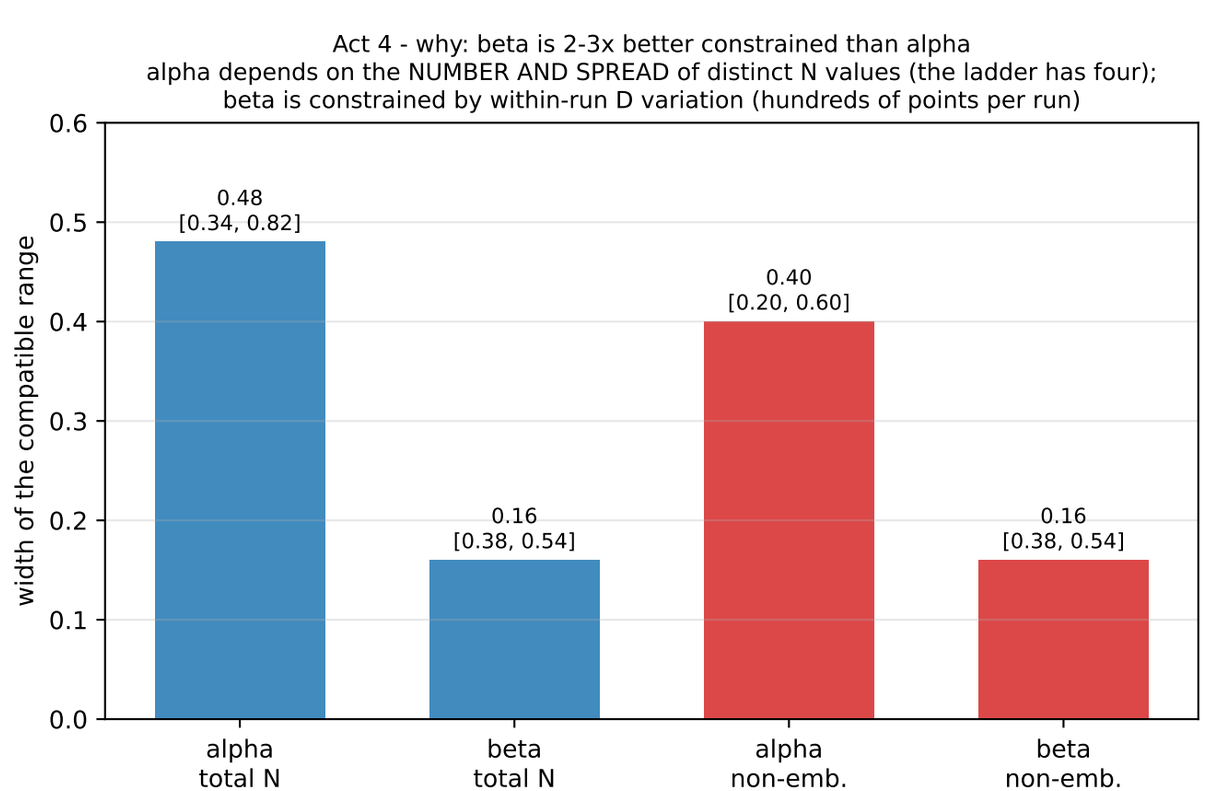

Act 4 - the profile likelihood over ALPHA
(dotted: each curve's own best fit minus 0.001 in R^2)
R^2 (against a constant-loss baseline)
alpha
total-parameter N (the paper's axis)
non-embedding N
published alpha = 0.34 (dash-dot line)
total N: compatible [0.34, 0.82] (width 0.48)
contains 0.34 AND 0.50: no discrimination


Act 4 - the profile likelihood over BETA
(the published 0.28 falls OUTSIDE the compatible range)
R^2 (against a constant-loss baseline)
beta
total-parameter N (the paper's axis)
non-embedding N
published beta = 0.28 (dash-dot line)
total N: compatible [0.38, 0.54] (width 0.16)
EXCLUDES Chinchilla's 0.28


Act 4 - the four leave-one-rung-out estimates, quoted individually
(dotted: each convention's full-fit point estimate)
fold-wise alpha
total-N compatible range
Chinchilla 0.34
total N
non-embedding N
0.492
0.580
0.501
0.517
0.415
0.422
0.398
0.408
0.421
0.333
0.344
hold out 0.3M
hold out 0.4M
hold out 0.8M
hold out 1.5M
hold out 2.6M
hold out 4.1M
hold out 6.3M
hold out 9.0M


Act 4 - why: beta is 2-3x better constrained than alpha
alpha depends on the NUMBER AND SPREAD of distinct N values (the ladder has four);
beta is constrained by within-run D variation (hundreds of points per run)
width of the compatible range
0.48
[0.34, 0.82]
0.16
[0.38, 0.54]
0.40
[0.20, 0.60]
0.16
[0.38, 0.54]
alpha total N
beta total N
alpha non-emb.
beta non-emb.